Assessment	Assignment 1		Total Marks	20
Course	CSE350		Bonus Marks	3
Section	05		Start	27 February 2026
Faculty	[ZAZ] Ihzaz Abrar Sadik		Deadline	**14 March 2026**
Semester	Spring 2026		Obtained Marks	out of **20**

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ID:		**Name:**	

Q1:

Let an analog signal have values in the range [-4(n+1)+2, 4(n+1)+2]. Where, 'n' is the last digit of your student ID. This signal is converted to a (mod((n+1), 2) +2)-bit digital signal using the fastest ADC available. This digital signal is then passed to a DAC of the same number of bits. The DAC has doubled resistances at its input branches.

a) What type of ADC & DAC were used? What is the bit number? [4]

b) Make a table listing the quantization intervals & the corresponding outputs from the DAC. [6]

c) If the input to the ADC is 'n+1', find the output at the DAC end. [3]

d) If you take $R = R_F$ for the DAC, find its reference voltage(s). [6]

e) If another signal $(n+1) + (n+1)\,sin(\omega t)$ of frequency $\frac{n+1}{4} kHz$ is passed as the input to the same ADC-DAC combination, and sampling is done at a rate of $(n+1)\,kHz$, find the output from the DAC for the first 3 samples. [4]